VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

PROGRAMMING FOR AI : ASSIGNMENT 1 REPORT

EXPLORATORY DATA ANALYSIS

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1 Exploratory Data Analysis: Tabular Data

1.1 Dataset Overview

For the tabular modality, we utilized the **Titanic Dataset** (Version: Yasserh/V1), sourced from Kaggle. This dataset is a cleaned version of the original Titanic competition data, focusing on passenger survival factors.

- **Source:** Titanic Dataset by Yasser Hesham

Dataset Statistics:

- **Total Samples:** 891
- **Features:** 12 (PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked)
- **Target Variable:** `Survived` (Binary: 0 or 1)

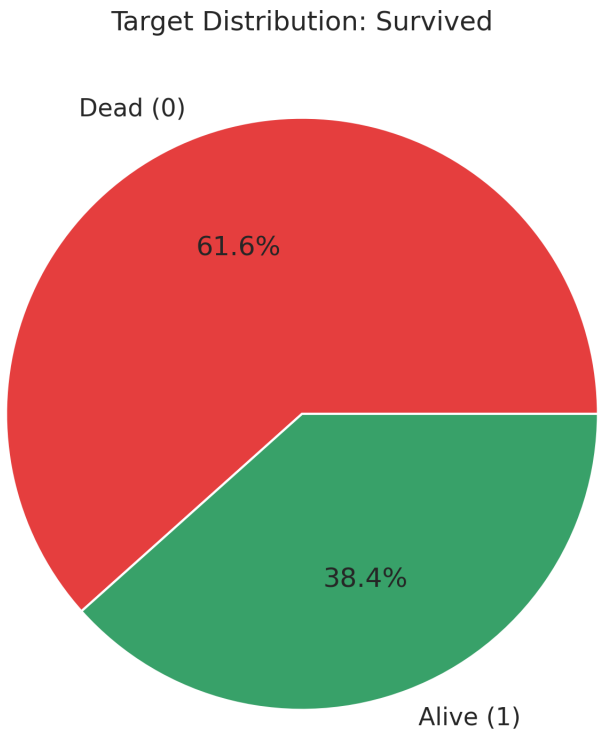

Figure 1.1: Target Distribution (Survived vs Dead)

1.2 Data Inspection and Missing Values

Upon initial inspection, we identified the following distribution of missing values:

- **Cabin:** 687 missing values (significant sparsity).
- **Age:** 177 missing values (requires imputation).
- **Embarked:** 2 missing values.

1.3 Key Findings

- **Survival Rate:** The mean survival rate is approximately **38.38%**.
- **Passenger Class:** The average `Pclass` is 2.3, suggesting a higher concentration of passengers in the 3rd class.
- **Age Distribution:** The mean age is roughly 29.7 years, with a standard deviation of 14.5. The youngest passenger was 0.42 years and the oldest was 80.
- **Fare:** There is a high variance in fares, ranging from 0 to over 512, indicating significant wealth disparity among passengers.

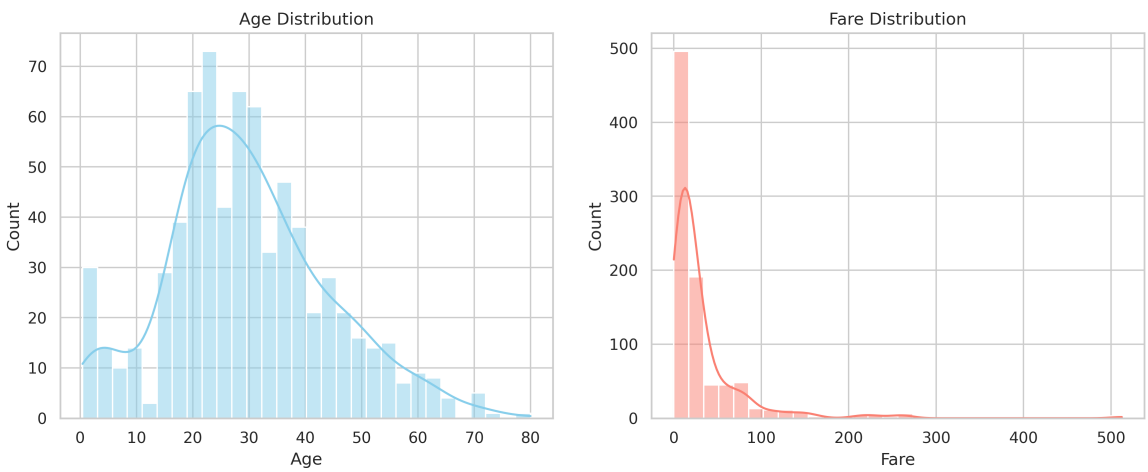

Figure 1.2: Distribution of continuous features: Age and Fare

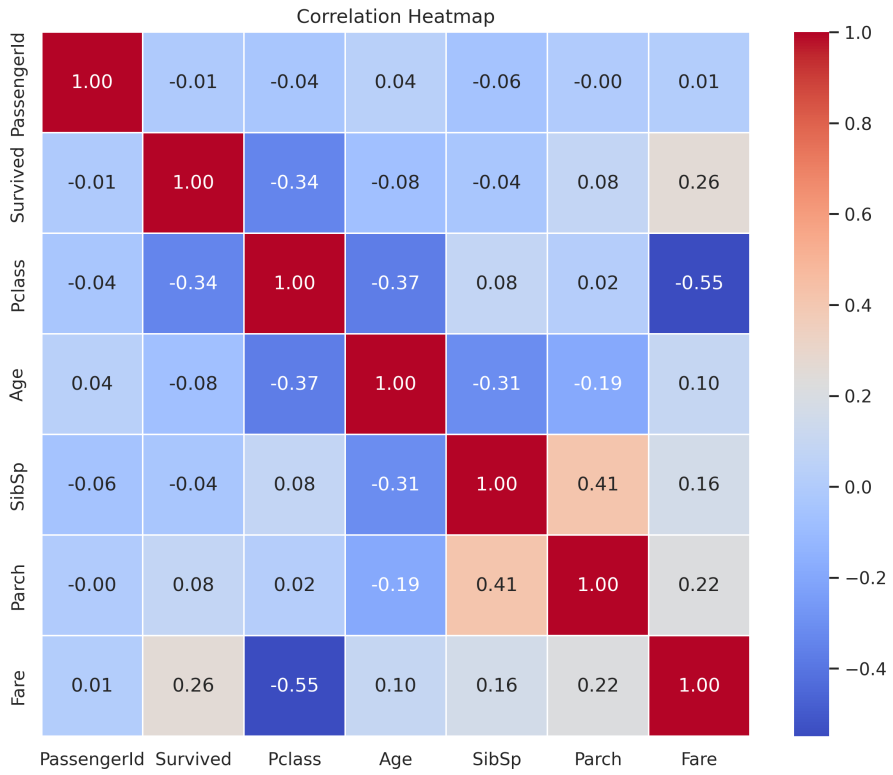

Figure 1.3: Pearson Correlation Matrix

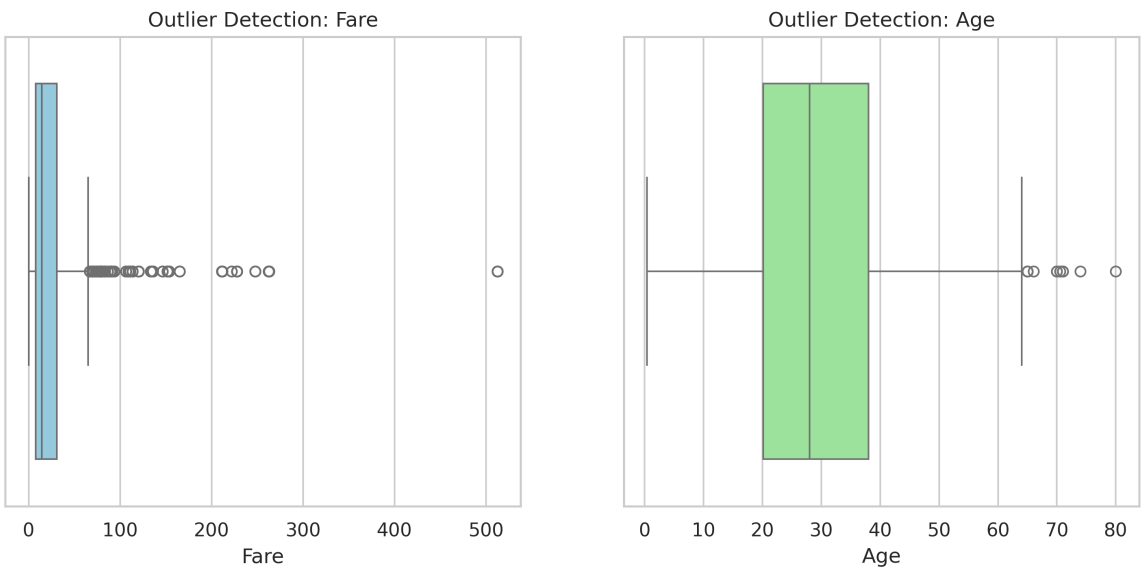

Figure 1.4: Outlier Detection via Boxplots

2 Exploratory Data Analysis: Text Data

2.1 Dataset Selection

For the text modality, we analyzed the **SMS Spam Collection Dataset**. This dataset consists of a set of SMS messages in English, tagged as being either "ham" (legitimate) or "spam".

- **Source:** SMS Spam Dataset by Marslino Edward

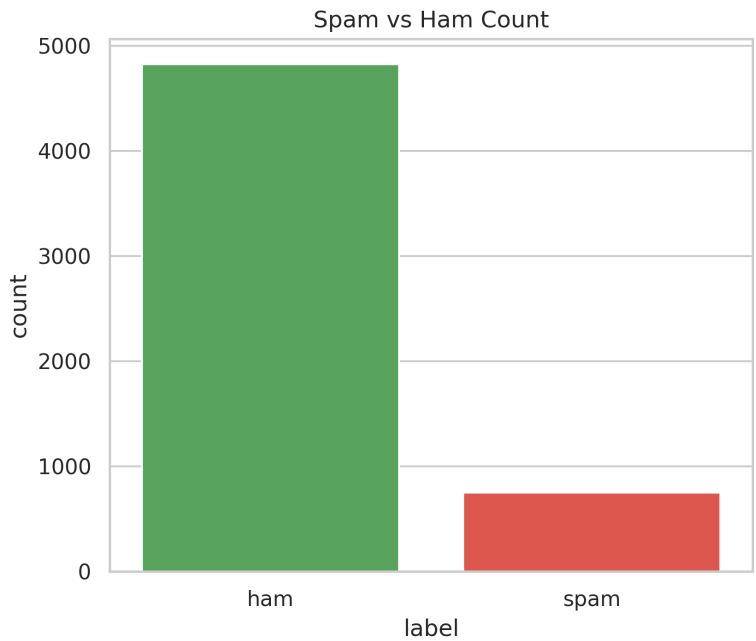

Figure 2.1: Frequency of Spam vs Ham Labels

2.2 Visualizations and Insights

To interpret the textual content, we utilized:

- **Word Clouds:** To visualize the most frequent terms after removing stop words.
- **Sequence Length Analysis:** Measuring the distribution of character and word counts to identify potential noise or outliers.

Note: The text analysis focuses on preparing the tokens for downstream NLP tasks, ensuring that the vocabulary size is manageable and identifying domain-specific keywords.

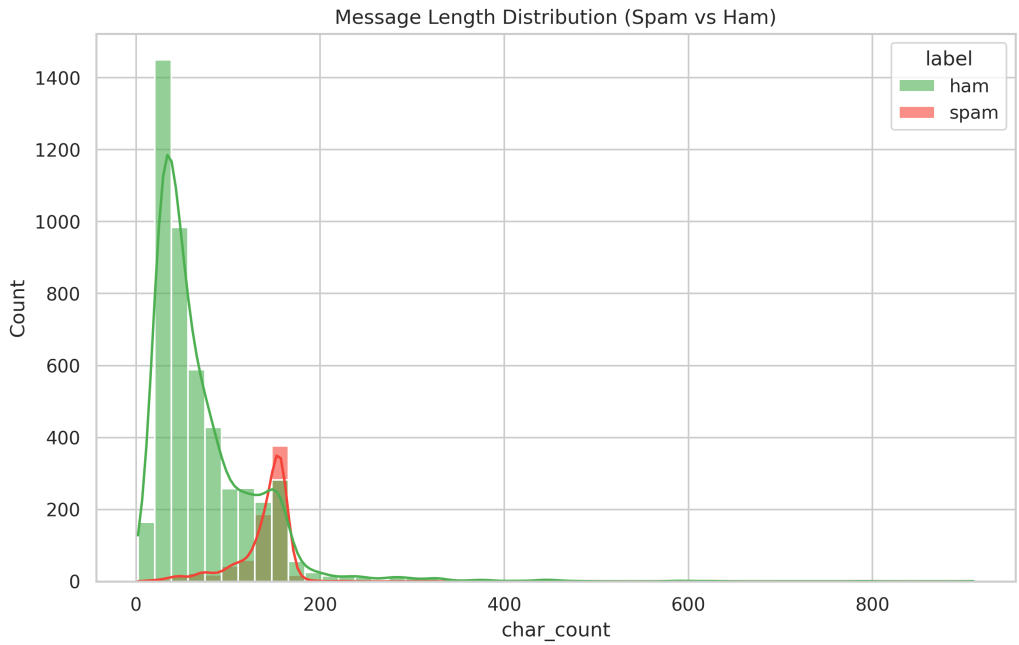

Figure 2.2: Message Length Distribution

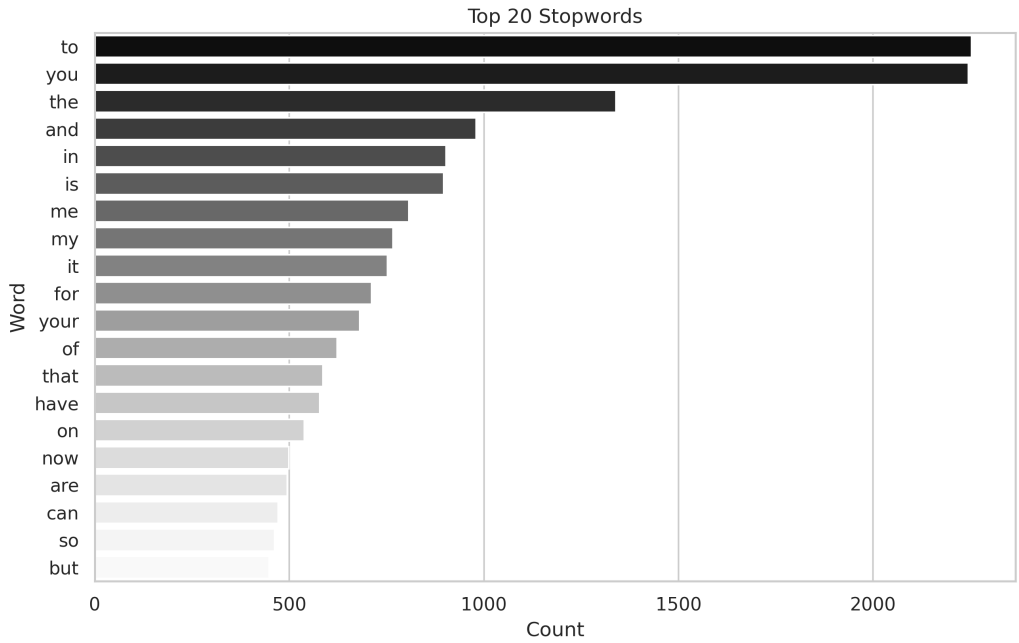

Figure 2.3: Top 10 Most Frequent Stopwords

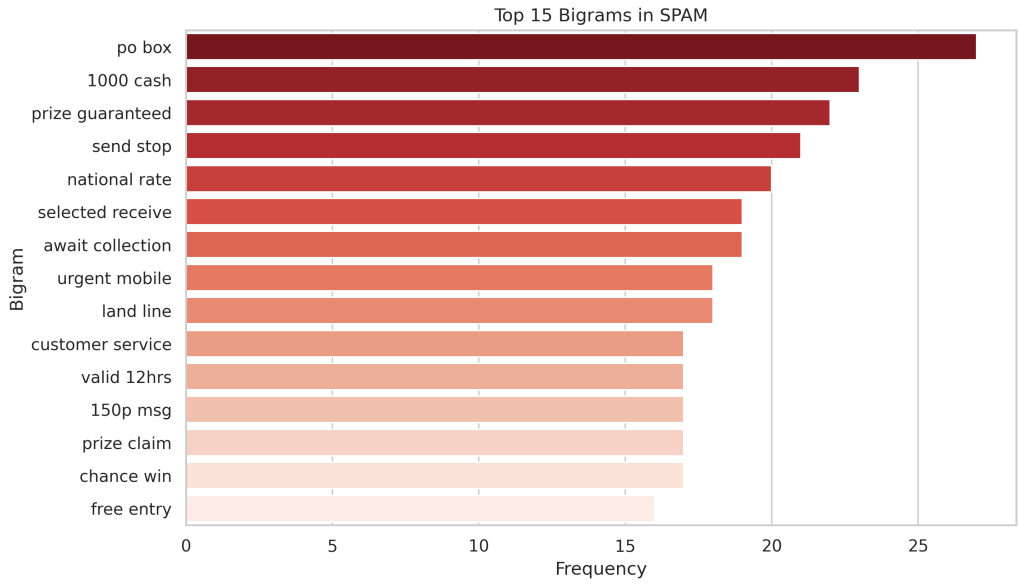

Figure 2.4: Top Bigrams in Spam Messages

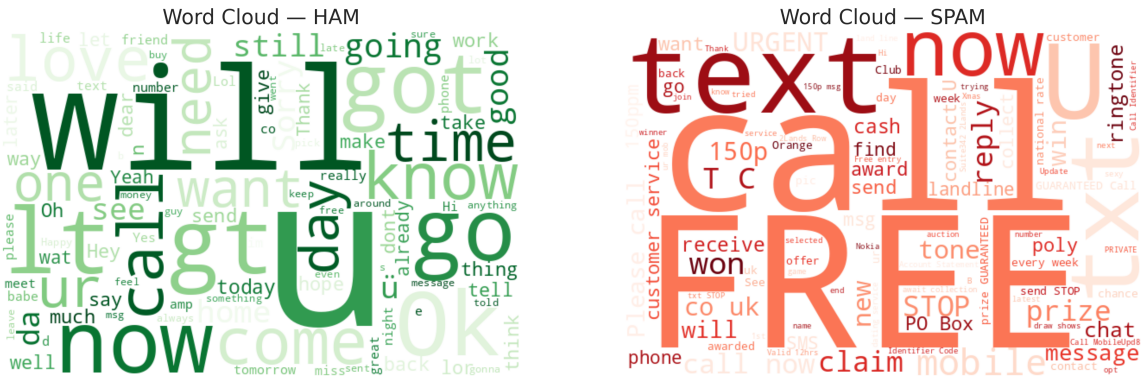

Figure 2.5: Word Clouds visualizing vocabulary dominance

3 Exploratory Data Analysis: Image Data

3.1 Dataset Overview

We utilized the **CIFAR-10** dataset (Python version), a staple for benchmarking image classification models developed by the Canadian Institute for Advanced Research.

- **Source:** Official CIFAR-10 Repository
- **Format:** 32×32 pixels, RGB color images.
- **Volume:** 50,000 training samples.

- **Classes:** 10 distinct categories (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck).

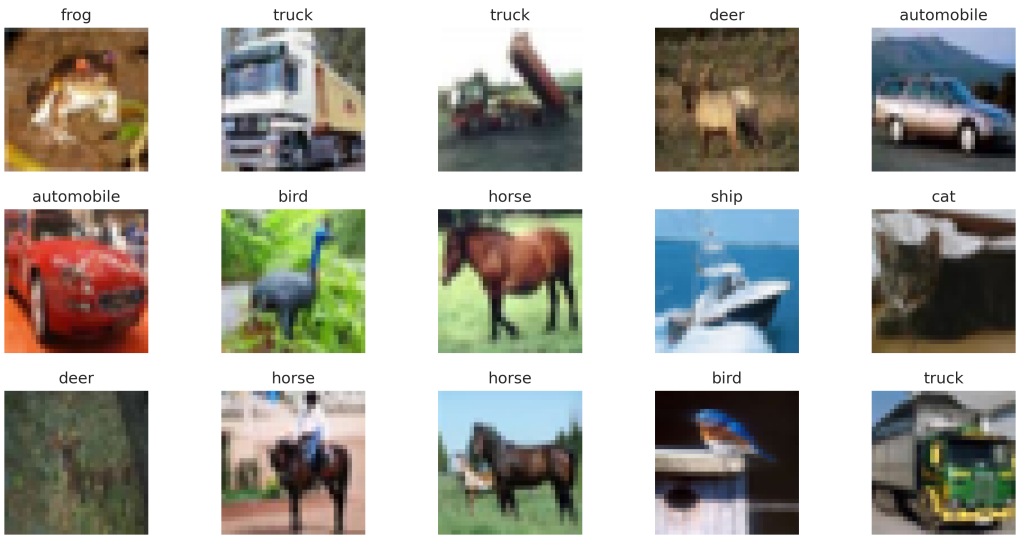

Figure 3.1: CIFAR-10 Random Image Samples

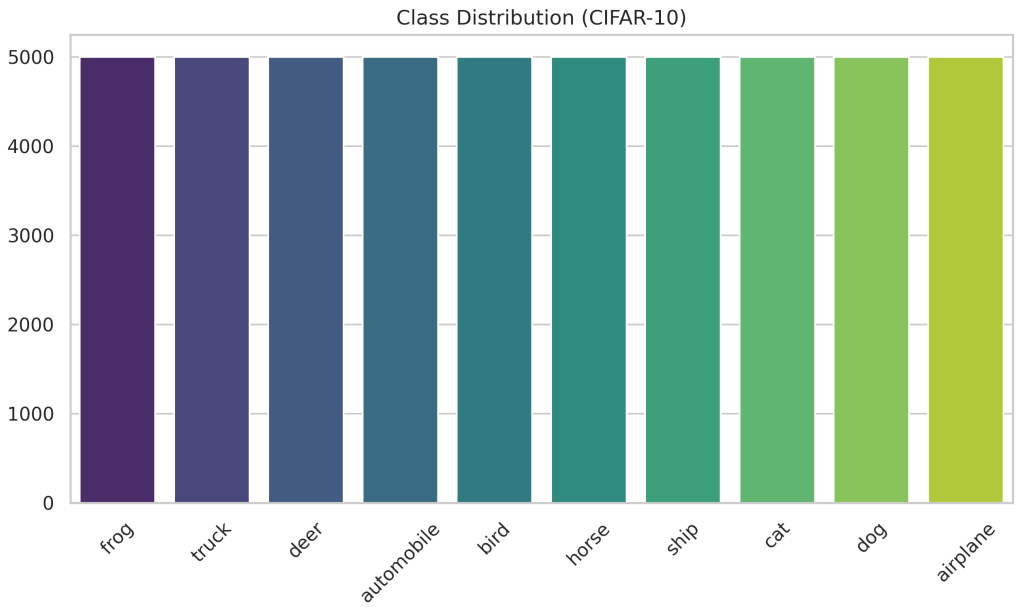

Figure 3.2: Class Distribution Across Dataset

3.2 Statistical Analysis of Pixels

We computed the mean and standard deviation for each color channel to guide future normalization steps:

Table 3.1: CIFAR-10 Channel Statistics

Channel	Mean	Std Deviation
Red	∼125.31	∼62.99
Green	∼122.95	∼61.90
Blue	∼113.87	∼66.56

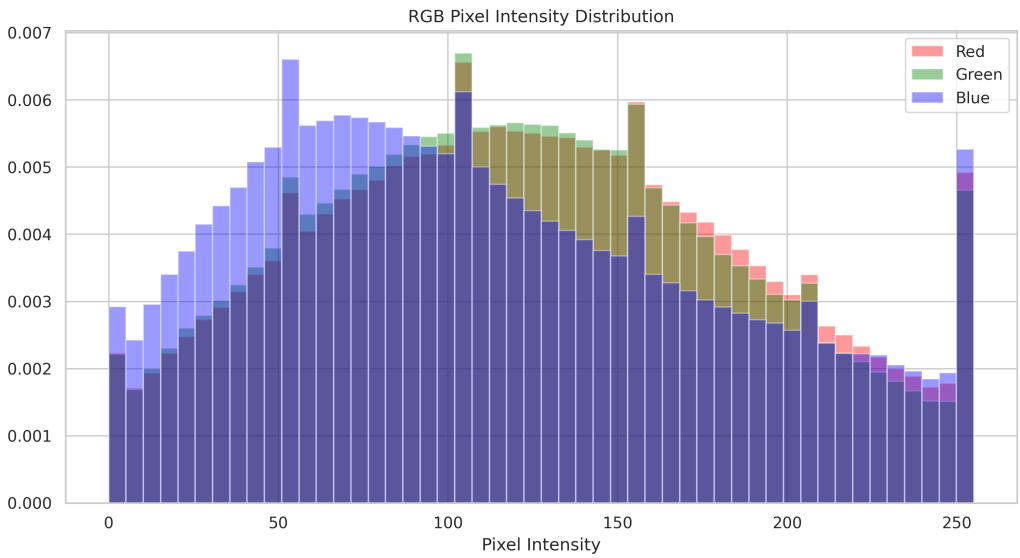

Figure 3.3: RGB Channel Intensity Profile

3.3 Key Findings

- **Class Balance:** The dataset is **perfectly balanced** with exactly 5,000 images per class, eliminating the need for oversampling or weighted loss functions.
- **Color Bias:** The Blue channel mean is lower than Red and Green, indicating an overall warmer tone in the dataset.
- **Variance:** A high standard deviation across all channels reflects a significant diversity in brightness and contrast levels.

4 Exploratory Data Analysis: Multimodal Data

4.1 Dataset Selection

For the multimodal modality, we explored the **Flickr8k Dataset**, a standard benchmark for image captioning tasks that pairs real-world images with descriptive natural language captions.

- **Source:** Flickr8k Dataset by Aditya Jain

Dataset Statistics:

- **Total Images:** 8,092 unique images.
- **Total Captions:** 40,460 (5 distinct human-annotated captions per image).
- **Format:** Varying image aspect ratios paired with English text sequences.

4.2 Data Inspection and Quality

An inspection of the text and image mapping reveals high data integrity:

- **Missing Values:** 0 missing captions or image references.
- **Image Dimensions:** Unlike CIFAR-10, Flickr images do not have a uniform resolution. Most images are scaled to have their longest edge at exactly 500 pixels (e.g., 500×333 or 375×500).

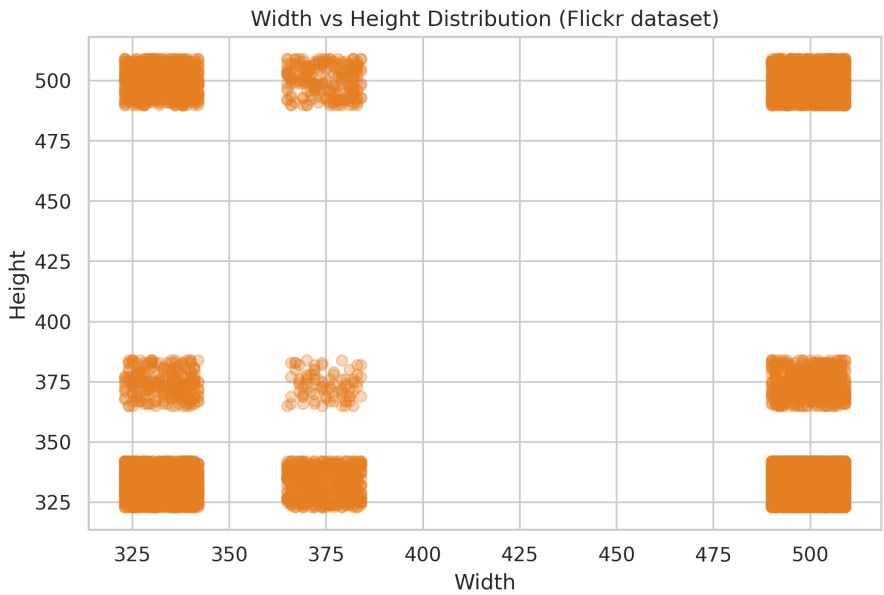

Figure 4.1: Image Resolution (Width vs Height) Distribution

4.3 Key Findings

- **Caption Length:** The vast majority of captions strictly range between 8 to 14 words. The lack of extreme outliers (very few sequences < 4 or > 30 words) makes sequence padding highly memory-efficient for NLP models.

- **Vocabulary Focus:** Nouns representing physical entities (*man, dog, boy, water, people*) dominate the frequency charts. The text is strictly focused on describing tangible subjects and actions rather than abstract concepts.
- **Preprocessing Need:** The varying image dimensions dictate that spatial standardizing techniques (such as center-cropping or zero-padding) must be applied before feeding the visual data into Convolutional Neural Networks.

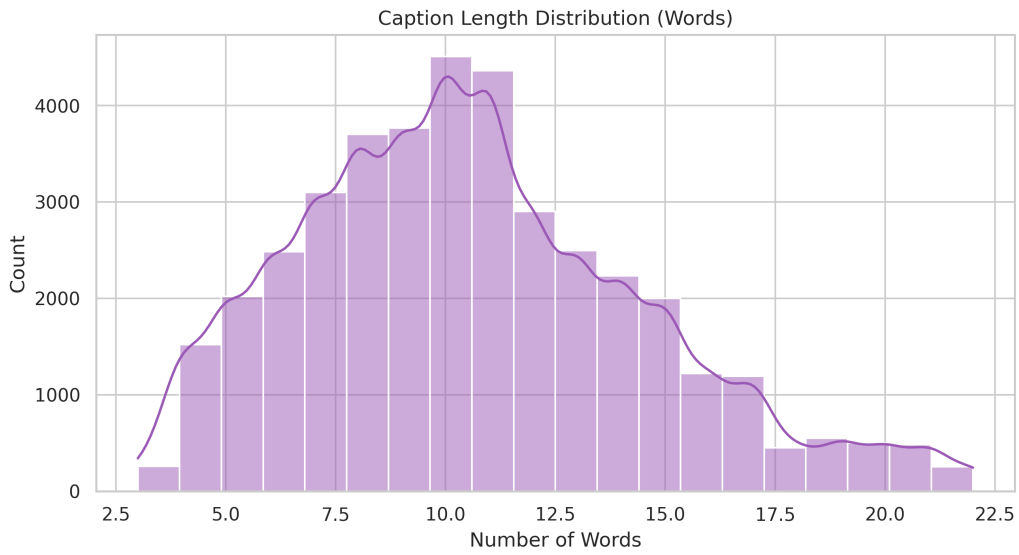

Figure 4.2: Caption Word Count Distribution

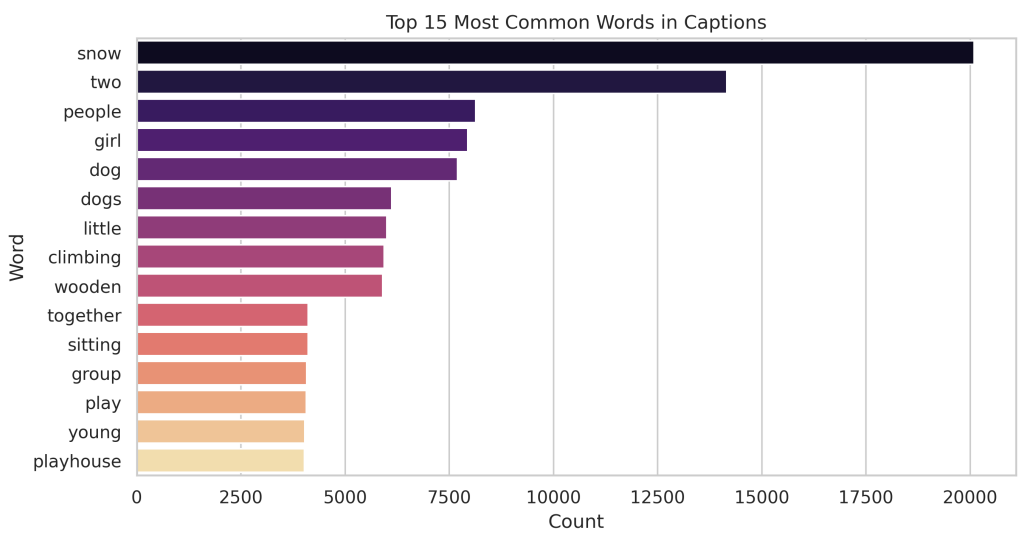

Figure 4.3: Top 15 Common Words in Captions

5 Multimodal Data Exploration (Flickr8k)

This section explores the **Flickr8k** dataset, a standard benchmark for multimodal tasks such as image captioning. The dataset pairs real-world images with descriptive natural language captions, allowing us to analyze the structural relationship between visual and textual data.

5.1 Dataset Overview

The Flickr8k dataset consists of a diverse set of images, each annotated with 5 distinct captions provided by human annotators.

- **Unique Images:** 8,092
- **Total Captions:** 40,460
- **Ratio:** 5 Captions / Image

5.2 Data Quality & Missing Values

An inspection of the text and image mapping reveals high data integrity:

- **Missing Values:** 0 missing captions or image references.
- **Duplicates:** No duplicate image-caption pairs were found.

5.3 Caption Text Analysis

The textual component of the dataset is highly constrained and descriptive.

- **Length Distribution:** The vast majority of captions range between 8 to 14 words.
- **Outliers:** There are very few extremely short (< 4 words) or extremely long (> 30 words) captions. This tight distribution is highly beneficial for training sequential models (like LSTMs or Transformers), as it minimizes the need for excessive padding and optimizes memory usage.

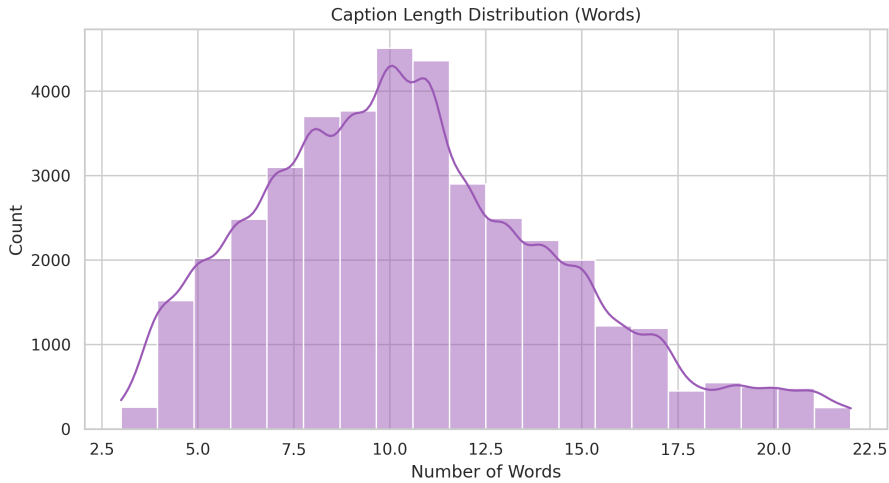

Figure 5.1: Distribution of Caption Word Counts

5.4 Vocabulary & Word Focus

After applying tokenization and removing standard English stopwords (e.g., 'a', 'the', 'in', 'on'), we analyzed the core vocabulary.

- **Dominant Tokens:** Nouns representing physical entities (e.g., *man, dog, boy, water, people*) and action verbs (e.g., *playing, running*) dominate the frequency charts.
- **Insight:** This vocabulary profile reflects the nature of image captioning tasks—the text is strictly focused on describing tangible subjects, actions, and environments rather than abstract concepts or sentiments.

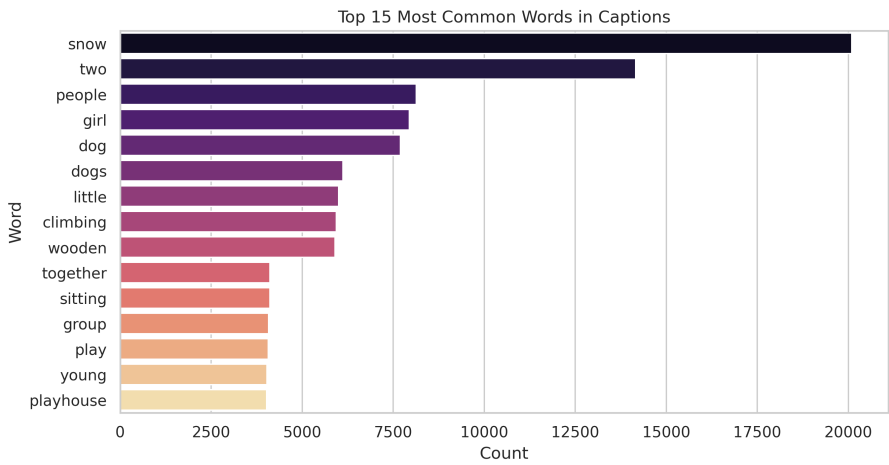

Figure 5.2: Top 15 Most Frequent Words (Excluding Stopwords)

5.5 Image Quality & Dimensions

Unlike standard computer vision datasets (e.g., CIFAR-10) which have uniform resolutions, Flickr images vary significantly in aspect ratios.

- **Resolution Diversity:** Images are typically scaled so that their longest edge is 500 pixels. The varying dimensions (e.g., 500x333 or 375x500) imply that standardizing techniques—such as center-cropping or zero-padding—must be applied before feeding the images into deep Convolutional Neural Networks (e.g., ResNet, VGG16).
- **Color Distribution:** The dataset exhibits a relatively balanced RGB intensity distribution, providing rich color contexts necessary for robust feature extraction.

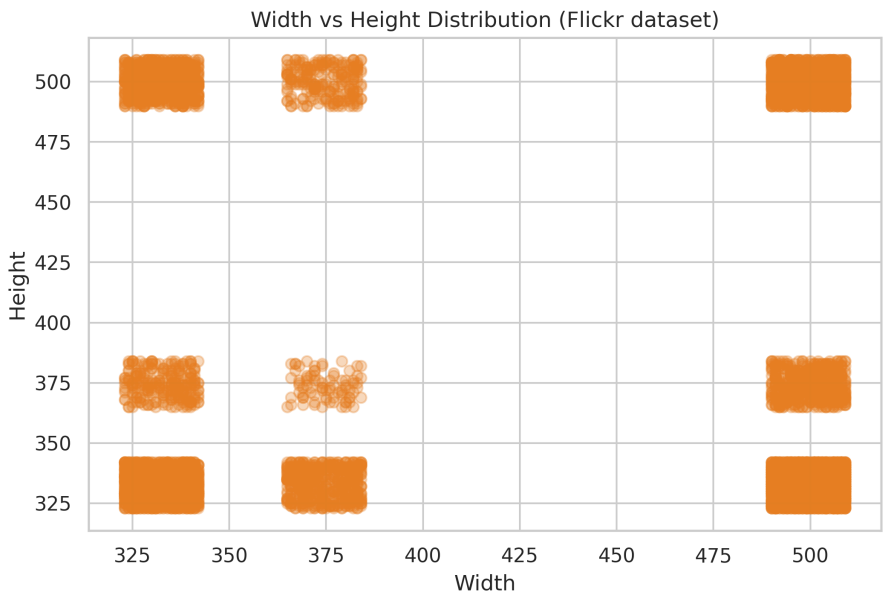

Figure 5.3: Image Resolution (Width vs Height) Distribution

Project Links & Resources

This report is accompanied by a fully interactive web-based EDA Report, open-source code, and a presentation video. Please visit the following links for full details:

- **Interactive Landing Page:** https://dangkhoi-dev.github.io/P4AI-DS/
- **Source Code Repository:** https://github.com/dangkhoi-dev/P4AI-DS
- **Presentation Video (YouTube):** https://youtu.be/A8Z70vnC_fE

6 Attribution & Data Sources

In compliance with data usage policies and academic integrity, we would like to acknowledge the original authors, creators, and curators of the datasets used in this Exploratory Data Analysis report. The datasets were sourced from the following repositories:

- **Titanic Dataset (Tabular Data):**
  Sourced from Kaggle. This dataset provides passenger information and survival status from the Titanic disaster.
  Link: https://www.kaggle.com/datasets/yasserh/titanic-dataset
- **SMS Spam Collection (Text Data):**
  Sourced from Kaggle. A public set of SMS labeled messages that have been collected for mobile phone spam research.
  Link: https://www.kaggle.com/datasets/marslinoedward/sms-spam-dataset/data
- **CIFAR-10 Dataset (Image Data):**
  Collected by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton. The dataset consists of 60,000 32x32 color images in 10 classes.
  Link: https://www.cs.toronto.edu/~kriz/cifar.html
- **Flickr8k Dataset (Multimodal Data):**
  Sourced from Kaggle. A standard benchmark dataset containing 8,000 images, each paired with 5 different human-annotated captions.
  Link: https://www.kaggle.com/datasets/adityajn105/flickr8k